

Unit 14, Lesson H2: Interpreting Relative Frequency Tables and Segmented Bar Graphs



Benchmarks

MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.

MA.912.DP.3.3 Given a two-way relative frequency table or segmented bar graph summarizing categorical bivariate data, interpret joint, marginal, and conditional relative frequencies in terms of a real-world context.

MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.



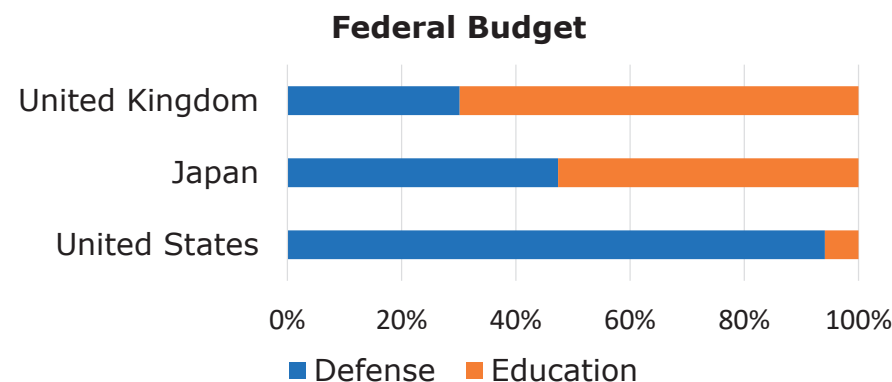
Learning Targets

- ☐ I can construct a two-way relative frequency table.
- ☐ I can interpret joint, marginal, and conditional relative frequencies in a real-world context.



14.H2.1 Warm-Up: Notice and Wonder

The segmented bar graph represents the distribution of where some of the 2009 federal budgets were expected to go.



1. What do you notice?
2. What do you wonder?



14.H2.2 Collaboration: Two-Way Relative Frequency Tables

Data was collected from a 2019 random sample of traffic accidents in Chicago. Data on the road conditions at the time of the accidents (dry, wet/slushy, icy) and the amount of damage caused (\$500 or less, \$501–\$1,500, more than \$1,500) is summarized.

- 27.76% of the accidents involved damage of \$500 or less.
- 28.77% of the accidents involved damage of \$501 to \$1,500.
- 78.43% of the accidents involved dry road conditions.
- 20.40% of the accidents involved wet/slushy road conditions.
- Of the accidents with damage of more than \$1,500, 69.77% involved dry road conditions.
- Of the accidents with damage of \$501–\$1,500, 80.22% involved dry road conditions.
- 24.79% of the accidents with icy road conditions had damage of \$501–\$1,500.
- 10.26% of the accidents with icy road conditions had damage of \$500 or less.

1. Use the data provided to construct a two-way relative frequency table. Round values to the nearest hundredth.

Traffic Accident Data

Road Conditions	Damage			Total
	\$500 or less	\$501–\$1,500	More than \$1,500	
Dry				
Wet/Slushy				
Icy				
Total				

2. Interpret the joint relative frequency for dry road conditions with damage of \$500 or less.
3. Interpret the marginal relative frequency for traffic accidents that had icy road conditions.

4. When the road conditions are icy, is the likelihood of the different amounts of damage the same? Explain your reasoning using conditional relative frequencies.

5. Does there appear to be an association between road conditions and the amount of damage caused?

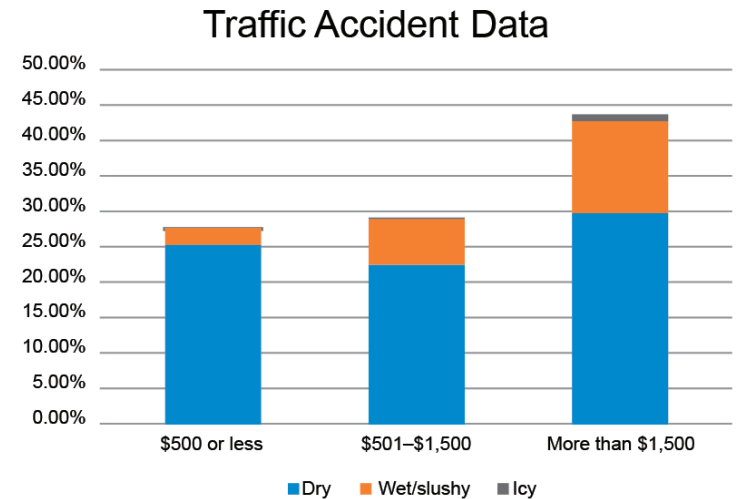
6. Does this mean that more hazardous road conditions cause more damage in traffic accidents?



Association does not imply causation!
If there is an association between two variables, it does not mean that one causes the other.
Two variables can be associated without having a causal relationship.

Another way to represent bivariate categorical data from a two-way relative frequency table is with a segmented bar graph. These graphs are helpful in visualizing comparisons between different variables by identifying how much larger or smaller different parts of each bar are compared to other bars.

The data from the two-way frequency table can be displayed in a segmented bar graph as shown.



7. With your partner, discuss the similarities and differences between the two-way relative frequency table and the segmented bar graph. Summarize your discussion.



14.H2.3 Try It: Interpreting Relative Frequencies

The General Social Survey (GSS) interviews a random sample of Americans every two years. In one question, respondents are asked to rate their general happiness level. Sample results for 1972, 1998, and 2021 are summarized in the table.

	1972	1998	2021	Total
Very Happy	486	891	783	2,160
Pretty Happy	855	1,575	2,308	4,738
Not Too Happy	265	340	923	1,528
Total	1,606	2,806	4,014	8,426

1. After reviewing the two-way frequency table, Diana said, "It seems like people in 2021 were a lot happier than in 1972."
 - a. Why do you think Diana said this?
 - b. Explain why this may not be an accurate conclusion to draw from the data.

2. Use the data to create a two-way relative frequency table. Round values to the nearest tenth of a percent.

	1972	1998	2021	Total
Very Happy				
Pretty Happy				
Not Too Happy				
Total				

3. Interpret the value 9.3% in context of the data.
4. Interpret the value 33.3% in context of the data.
5. Interpret the value 25.6% in context of the data.

6. Interpret the value $\frac{27.4\%}{47.6\%}$ in context of the data.

7. What percentage of the respondents from 1972 reported they were very happy?

8. What percentage of the respondents from 1998 reported they were very happy?

9. What percentage of the respondents from 2021 reported they were very happy?

10. What can be concluded about the general happiness of Americans from the two-way relative frequency table?



14.H2.4 Wrap-Up: Ticket Out the Door

1. The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City.

The two-way relative frequency table classifies the passengers as whether the person survived (Yes or No) and the class of the ticket they had (1st, 2nd, 3rd, or Crew).

Ticket	Survived		Total
	No	Yes	
1st	5.5%	9.2%	14.8%
2nd	7.6%	5.4%	12.9%
3rd	24.0%	8.1%	32.1%
Crew	30.6%	9.6%	40.2%
Total	67.7%	32.3%	100.0%

a. What does the value 30.6% mean in context of the data?

b. What does the value $\frac{30.6\%}{40.2\%}$ mean in context of the data?